STUDIES ON A SERIES OF TRYPTOPHAN-INDEPENDENT STRAINS DERIVED FROM A TRYPTOPHAN-REQUIRING MUTANT OF ESCHERICHIA COLI[1]

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STRAINS which approach wild type in appearance or behavior can arise from mutant strains as a result of either reversion (mutation at the original mutant locus) or suppression (mutation at some other genetic locus). In microorganisms both reversions (HOGNESS and MITCHELL 1954) and some suppressor mutations (YANOFSKY 1952; PARKS and DOUGLAS 1957) are known to act by restoring the specific enzyme activity which was lacking in the original mutants examined. Although it might be expected that reversion would always restore a mutant locus to its wild type form, recent studies with Neurospora indicate that this may not always be true. Both GILES (1957) and FINCHAM (1957) have reported cases in which reversion restored somewhat less than the wild type level of specific enzymes.

The purpose of the study reported in this paper was to examine the characteristics of the different types of strains which were obtained by mutation of a tryptophan- requiring mutant of *Escherichia coli* to tryptophan independence.

Terminology

The term "phenotypic revertant" is used to describe any tryptophan-independent (T^+) strain, regardless of genetic type, derived from the tryptophan- requiring mutant td_2. The term "reversion" or "revertant" will apply only to strains in which the restorative mutation appears to have occurred at the same locus as the original mutation.

MATERIALS AND METHODS

Escherichia coli strains: The tryptophan auxotroph from which the tryptophan-independent strains employed in this study were derived is designated td_2. This mutant was obtained from wild type strain K-12 by using ultraviolet light as the mutagenic agent, followed by penicillin selection. Nutritional and enzymatic studies with this auxotroph have shown that it will not respond to indole or anthranilic acid and specifically lacks tryptophan synthetase activity (the enzyme catalyzing the coupling of indole and serine to form tryptophan). When grown in liquid minimal medium with low tryptophan supplements (2–10 μg

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/ml), this strain accumulates large amounts of indole in its culture medium (YANOFSKY 1957).

In some experiments with strain td_2 it was desirable to have an unlinked marker present. For this purpose a histidine-requiring td_2 stock (td_2H^-) was prepared from an H^- cystine-dependent strain by transduction with bacteriophage grown on strain td_2. Since the cystine marker is closely linked to td_2, a large percentage of the cystine independent colonies obtained were td_2H^-. A stock bearing the same cystine marker (cys^-td^+) was used in all crosses and transductions.

Media: L-broth was used when a nutrient medium containing tryptophan was required (LENNOX, 1955), except in crosses, for which Difco Penassay broth was employed. The minimal salts medium of VOGEL and BONNER (1956) with 0.2 percent added glucose was used throughout and was supplemented with L-tryptophan when required. A supplement of 10 μg L-tryptophan/ml. permitted full growth of td_2. A supplement of 30 μg L-histidine/ml. was used when necessary.

Spontaneous reversion experiments: A 24–48 hour-old single colony isolated from strain td_2 was suspended in saline and the turbidity of the suspension determined. A series of L-broth tubes (10 ml/tube) were then inoculated so that each tube contained 10^4 bacteria/ml. At this level of dilution, tryptophan-independent cells would not be expected to be present. To verify this in each case, aliquots of the original suspension (10^5 cells) were plated on plates containing minimal agar plus 0.01 μg tryptophan/ml. The broth cultures were grown overnight with shaking and in the morning contained approximately 5×10^9 bacteria/ml. Saline dilutions were made from each L-broth tube and aliquots containing 10^7 cells were plated on plates containing minimal agar with the low level tryptophan supplement (0.01 μg/ml)[4].

Experimental and control plates were incubated for eight days at 37°. Although some tryptophan-independent colonies appear as early as one or two days after plating, the number of such colonies reaches a maximum after about eight days.

Reversion experiments with ultraviolet light: Strain td_2 was grown overnight in L-broth, and in the morning the culture was diluted to 10^8 bacteria/ml in a fresh tube of L-broth. This tube was incubated with shaking until a population of 5×10^8 cells/ml was obtained, and then this second culture was diluted to 10^7 bacteria/ml in minimal medium without glucose. Ten ml of this suspension were exposed to ultraviolet light at a distance of 40 cm. for ten seconds. With strain td_2 this exposure gives approximately 80–90 percent kill. Samples containing 10^7 cells were plated at zero time and after ten second exposure (10–20 percent viable) on minimal agar supplemented with 0.01 μg tryptophan/ml. The plates were incubated for eight days at 37°.

Routine handling of tryptophan-independent colonies: Colonies of widely varying sizes appeared after incubation of the experimental plates for eight days. These colonies were picked and streaked on minimal agar to obtain single colony

[4] This low tryptophan supplement will not support visible growth of td_2 but about ten times more ultraviolet-induced phenotypic revertants are recovered with this supplement than with minimal medium alone.

isolates before further tests were performed. Following this initial purification the T^+ strains were tested for indole accumulation and were plated to obtain an accurate determination of colony size.

Accumulation studies: Tubes containing minimal medium with 0.2 percent glucose or 0.2 percent glycerol were inoculated with the T^+ strains to be tested. These cultures were incubated with shaking for 40–48 hours at 37°. One part of the culture filtrate was mixed directly with two parts 95 percent ethyl alcohol and one part Ehrlich's reagent as a test for indole accumulation. Ehrlich's reagent gives a bright red color in the presence of trace amounts of indole.

Colony size studies: When isolates of different phenotypic revertants were first examined it was apparent that there was a wide range of individual growth rates. Accurate measurement of log phase generation time of a large number of these T^+ strains seemed impractical; consequently, colony size measurements were performed under controlled conditions. It was felt that such measurements would give a rough indication of the relative growth rate of any one mutant strain and would permit comparisons of the growth rates of the phenotypic revertants and of wild type. The log phase growth rate of selected T^+ strains was examined more carefully as described in the next section.

Single colony isolates from each T^+ strain were grown for 16 hours at 37° in nutrient medium. In this length of time each culture reached the stationary phase of growth and the cells were present at a concentration of about 5×10^9/ml. These cultures were diluted in saline and 10 to 30 cells were plated on minimal agar. The plates were incubated for 96 hours at 37°. At this time, the colonies were measured under a binocular microscope equipped with an ocular micrometer. The "average colony size" was obtained by taking the numerical average of the diameters of all the colonies on the plate. Colonies derived from some T^+ strains were quite uniform in size and showed a variation of only 0.1 mm or less between colonies. However, some strains consistently threw off fast-growing (large) colonies. These strains were usually the slowest growing isolates. In such cases, "average colony size" was not very meaningful and these strains were used in only a few of the experiments. In cases where it was desirable to use strains of this type there is an explanatory notation.

Experimental variation between duplicate plates in the same experiment and between experiments was determined for one T^+ strain having a relatively stable colony size. The variation observed was less than ten percent.

Growth curves: Log phase growth rates on liquid minimal medium were determined for selected T^+ strains. Erlenmeyer flasks (125 ml) equipped with sidearm colorimeter tubes were used in these studies. These flasks, containing 10 ml of medium, were inoculated to give an initial concentration of 10^8 bacteria/ml and were incubated with shaking in a 37° water bath. The turbidity of the cultures was measured with a Klett-Summerson photoelectric colorimeter (660 filter) at 30 minute intervals throughout lag phase and log phase.

Studies of enzymatic activity: Strains selected for enzyme analysis were grown in one liter volumes of minimal medium. Log phase cells were used as the starting inocula. Each culture was incubated with shaking at 37° and its growth was followed closely. When the turbidity indicated that the culture had reached a population of 5×10^8 to 10^9 cells/ml an aliquot was removed for the determination of colony size; the remaining cells were harvested by centrifugation. The cells were washed once with saline and resuspended in 0.1M phosphate buffer (pH 7.8). These suspensions were then disrupted in a 9 kc Raytheon sonic oscillator and the cellular debris removed by centrifugation for 30 minutes at 15,000 g in an SS-1 Servall centrifuge. The supernatant solutions were used as crude enzyme preparations and were stored at −15°C.

Tryptophan synthetase activity was measured by the method of YANOFSKY (1957). One unit of tryptophan synthetase is that amount of enzyme which will convert 0.1 uM of indole to tryptophan when incubated for 30 minutes at 37° in the presence of indole, serine and pyridoxal phosphate. The protein content of the extracts was determined by the method of LOWRY *et al.* (1951). The tryptophan synthetase specific activity of an extract is expressed in units of enzyme activity per milligram of extracted protein.

Genetic studies: Suppressed mutants and true revertants were distinguished by using two genetic techniques, transduction and crossing. Transductions were performed with phage Plkc using the procedure described by LENNOX (1955). All crosses were performed in the following way: An F^+K-12 strain requiring cystine ($cys^- td^+$) was grown overnight with shaking at 37° on Penassay broth plus added cysteine. Growth under these conditions presumably converts the F^+ character to an F^- phenocopy (LEDERBERG, CAVALLI and LEDERBERG 1952), and this strain was treated as the recipient. The tryptophan-independent strains obtained from the derived strain td_2H^- were used as donors. These strains were grown overnight without aeration. In all crosses, 10^9 recipient cells and 10^8 donor cells were added to 2 ml of Penassay broth. This mixture was incubated with shaking for one hour at 37° and samples were diluted and plated on minimal agar enriched with 0.2 percent nutrient broth[5] and supplemented with 10 μg of L-tryptophan/ml.

The genetic constitution of the progeny from both transductions and crosses was tested by replica plating (LEDERBERG and LEDERBERG, 1952) followed by picking and streaking on appropriate media.

RESULTS

Frequency of appearance of indole accumulators among phenotypic revertants: Accumulation tests were performed on purified isolates chosen at random from the phenotypic revertants derived spontaneously and induced by ultraviolet light to determine whether these strains differed from wild type. The tests showed that most of the T^+ strains obtained were accumulators of indole, and that the percentage of nonaccumulators was smaller in the ultraviolet treated material (Table

[5] The td_2H^- strain shows high frequency recombination for markers in the tryptophan region when the crossing mixture is plated on minimal medium enriched with 0.2 per cent nutrient broth (YANOFSKY and LENNOX, unpublished).

1). The work described in the Results section is entirely concerned with the behavior of the indole accumulators.

All of the ultraviolet-induced T^+ strains probably derive from independent mutational events, but under the conditions employed in the spontaneous reversion experiments some duplication of specific T^+ types could occur. Spontaneous T^+ strains must of course be independently derived if they appeared in separate experimental cultures or if they can be distinguished by characteristics of growth and enzyme content.

TABLE 1

Frequency of appearance of indole accumulators among tryptophan-independent mutants

	Ultraviolet		Spontaneous	
	Control (0 sec)	Experimental (10 sec)	Control	Experimental
Total cells plated (3 experiments)	1.2×10^8	4.6×10^7	3×10^5	2.8×10^8
Number of phenotypic revertants	22	363	0	45*
Frequency	$0.18/10^6$	$7.7/10^6$	0	$0.16/10^6$
Percent indole accumulators in random sample		90.9		54.0

* Including at least 22 independent events since this number of different experimental cultures were examined.

Variation in colony size among ultraviolet-induced and spontaneous phenotypic revertants: The indole accumulating strains employed in the colony size studies form an unselected sample. Twenty-nine spontaneous T^+ strains and 59 ultraviolet-induced T^+ strains were examined and the results are plotted in Figures 1a and 1b. These figures show that different patterns are obtained with the spontaneous and ultraviolet-induced populations. The spontaneous phenotypic revertants have average colony diameters which appear to be distributed randomly over a wide size range, 0.5 to 4.5 mm. Although a few of the ultraviolet-induced T^+ strains have colony diameters of 2.0 to 3.7 mm, most fall in the 0.8 to 1.7 mm class. There is probably another class at 0.5 mm.

This discrepancy between the two populations might possibly be explained by selection against slow growing spontaneous phenotypic revertants. In spontaneous reversion experiments, a phenotypic revertant must be able to grow and survive in a large population of td_2 cells which are growing rapidly on the nutrient medium supplied. This is not the case in the experiments employing ultraviolet light. In these, the mutations presumably occur at or soon after the time of plating on minimal agar and there is, therefore, no possibility of competitive growth on a nutrient medium. If there is selection against slower growing T^+ strains in spontaneous mutation experiments the frequency of their appearance in plating experiments will not represent the actual frequency of this type of mutation. Experiments performed to examine this point did not reveal any obvious selection against slow growing phenotypic revertants on the nutrient medium employed.

The correlation of log phase generation time with colony size: The colony size differences observed in the experiments previously described might not reflect differences in growth rate but could be due to differences in the length of the lag phase. To examine whether colony size is correlated with log phase generation time, the log phase generation time of five spontaneous phenotypic revertants

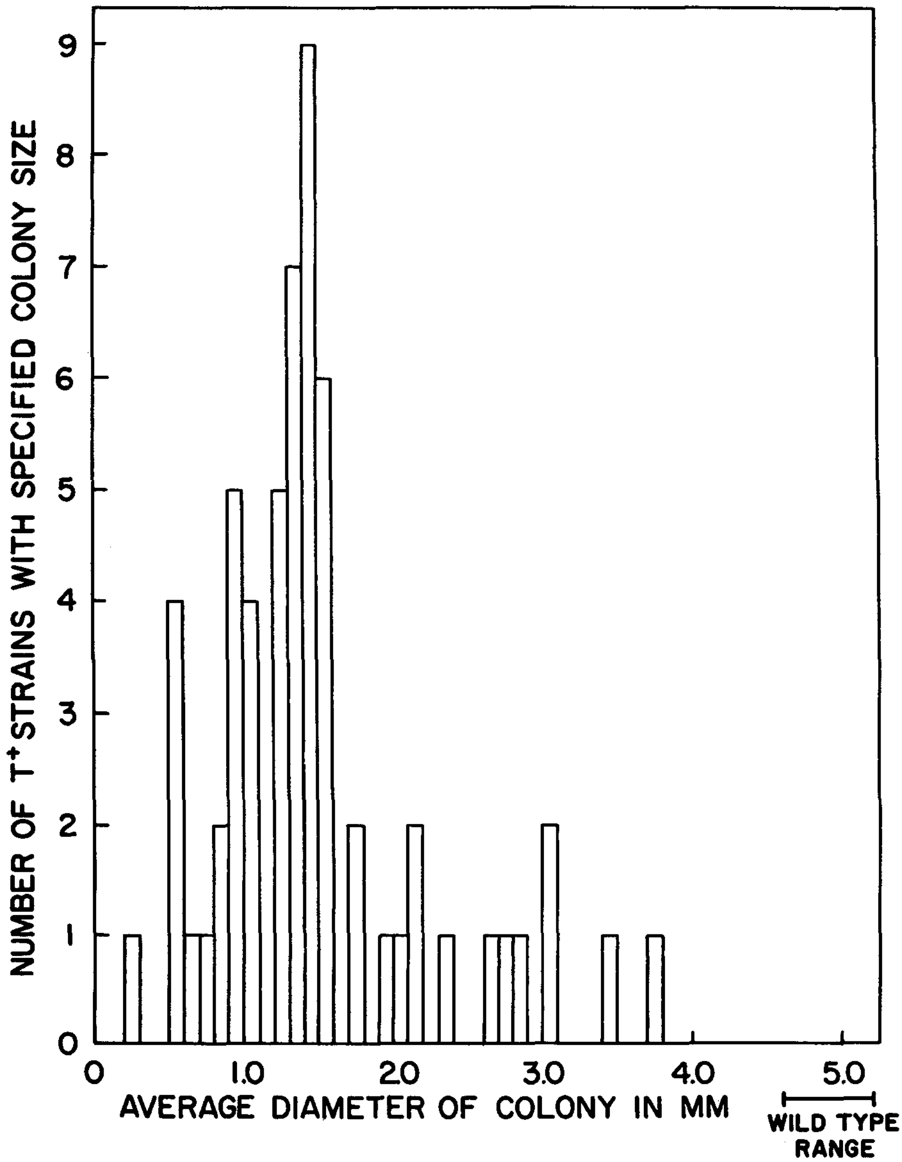


FIGURE 1a.—Average colony sizes in an unselected sample of indole accumulating ultraviolet-induced phenotypic revertants.

was determined and the values obtained plotted versus the colony sizes of the same strains (see Figure 2). It is clear from the figure that colony size appears to be a true reflection of the growth rate of these strains.

Genetic studies: Genetic studies were performed on a large number of indole-accumulating T^+ strains to determine whether the mutations conferring tryptophan-independence had occurred at the td_2 locus or at some suppressor locus.

Transductions were carried out in the following way. Phage was grown on the T^+ donor strains and the lysates obtained were mixed with cells of a cys^- td^+ recipient. Aliquots were plated on minimal agar supplemented with 10 μg tryp-

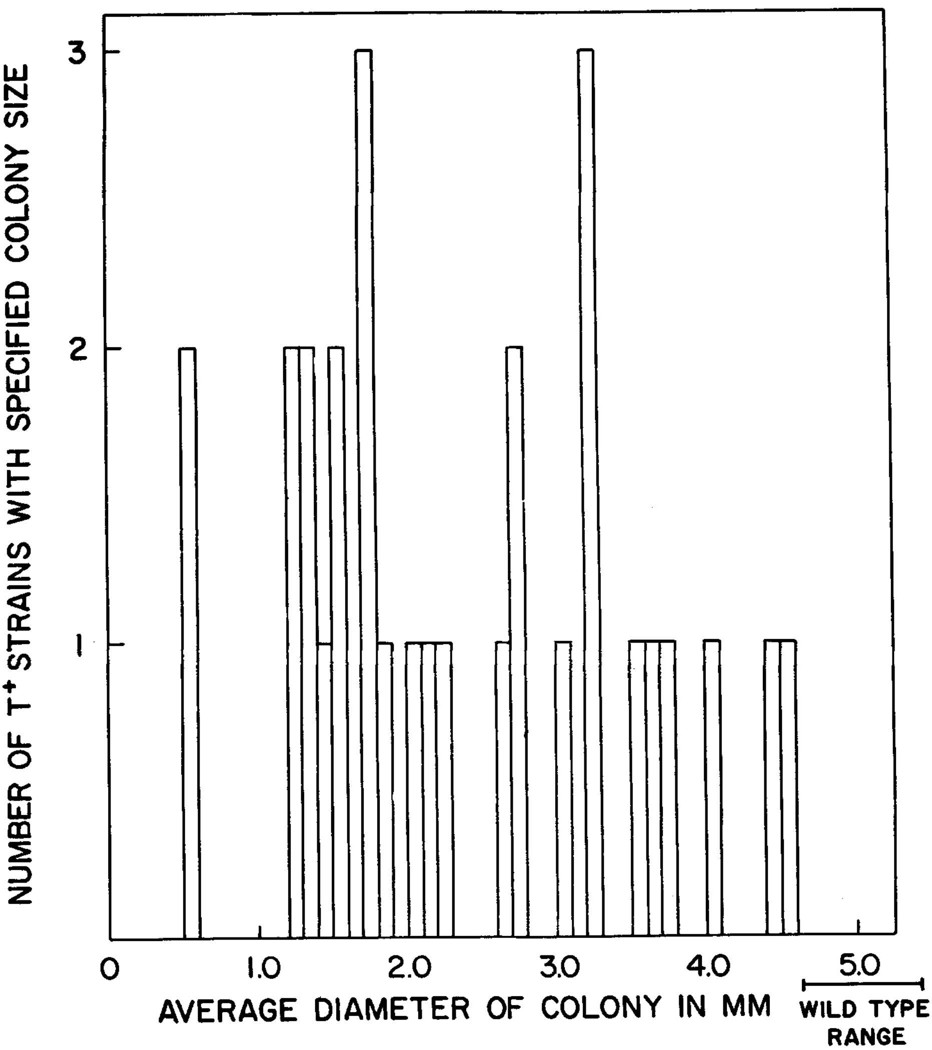


FIGURE 1b.—Average colony sizes in an unselected sample of indole accumulating spontaneous phenotypic revertants.

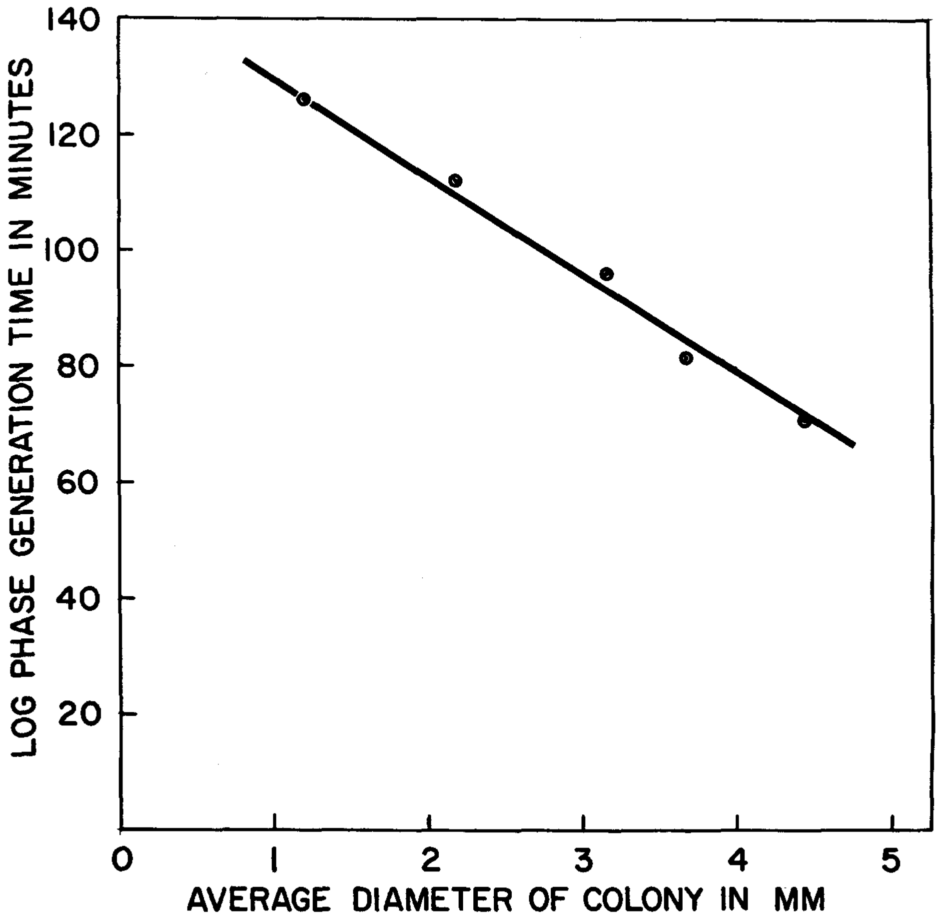


FIGURE 2.—Correlation of log phase generation time with colony size in spontaneous T+ strains selected for uniformity of colony size.

tophan/ml. thereby making the *cys*$^+$ marker selective and the tryptophan marker nonselective. The transduced colonies were replicated to minimal agar and, if tryptophan requirers were present, the number of T$^+$ and T$^-$ colonies were recorded. When they appeared, some tryptophan-requiring colonies were picked from each transduction and their nutritional requirement verified by streaking on appropriate media.

Since the *cys*$^-$ marker used in this experiment is linked to the td_2 locus and the *cys*$^+$ marker of the donor strain was selected, some of the transduced colonies should carry the donor form of the *td* locus. If the donor strain carried td_2 unchanged, plus an unlinked suppressor gene, some of the transduced colonies should be tryptophan-dependent. If the suppressor gene were completely unlinked, approximately 36 percent of the *cys*$^+$ colonies should require tryptophan (YANOFSKY and LENNOX, unpublished). This is the value obtained in control transductions of td_2 into the cys^- strain. If the suppressor gene carried by the donor strain showed some linkage to the td_2 locus, the number of tryptophan-dependent colonies would be reduced. On the other hand, if the tryptophan-independent donor strain were derived from a mutation at the td_2 locus, tryptophan-requiring colonies would not be expected.

Genetic data obtained in transductions with selected ultraviolet induced T^+ strains are presented in Table 2. Eight of the eleven phenotypic revertants tested

TABLE 2

Genetic data on selected ultraviolet-induced phenotypic revertants

Colony size	Genetic test	T^-/colonies tested	Percent tryptophan requirers
2.7	Transduction	0/152	0
2.6	Transduction	0/123	0
2.1	Transduction	0/321	0
1.3	Transduction	73/193	37.8
1.3	Transduction	12/59	20.3
1.3	Transduction	5/44	11.3
1.1	Transduction	105/304	34.6
0.9	Transduction	16/91	17.7
0.5	Transduction	56/165	33.9
0.5	Transduction	35/121	28.9
0.2	Transduction	10/97	10.3
	Control transduction ($td_2 \rightarrow cys^-$)	72/215	33.5

T^- = colonies requiring tryptophan.

proved to carry suppressors, while three appeared to represent true reversions of the td_2 locus.

Genetic data on the spontaneous phenotypic revertants were obtained by crossing as well as by transduction. In the crosses the recipient strain was also cys^- td^+. Spontaneous T^+H^- strains derived from strain td_2H^- were used as donors. After crossing, aliquots of dilutions were plated on enriched minimal agar supplemented with tryptophan. Thus, H^+ cys^+ progeny were selected and the tryptophan requirement was kept nonselective, as in the transduction experiments. Control crosses of cys^- by td_2H^- showed that when the cys^- gene was replaced by cys^+ from the donor, the td_2 gene was included about 32 percent of the time.

The rationale of the cross is the same as that of the transduction. If the donor strain contains td_2 and an unlinked suppressor gene, td_2 will be recovered in about 32 percent of the cys^+ progeny. If the donor strain carries a td_2 reversion, no tryptophan requirers will be recovered in the progeny of the cross.

In order to check a large number of phenotypic revertants, a mixed crossing technique was devised. In this procedure, instead of examining each phenotypic revertant singly, a mixture of several was employed as donor material. In control experiments of this type (see Table 3) it was shown that in a mixture of four strains, if one suppressed mutant was present, tryptophan-requiring colonies were recovered.

The results of genetic tests on selected spontaneous phenotypic revertants are presented in Table 4. Three of the 26 strains examined yielded tryptophan requirers and therefore carry suppressor mutations while the remaining 23 are true reversions. The results of tests on a random sample of spontaneous phenotypic revertants unselected for colony size are presented in Table 5. In this case all T^+ strains tested proved to be true reversions.

Correlation of genetic type with growth rate as determined by colony size: Most of the genetic studies were performed with spontaneous and ultraviolet-induced mutants selected on the basis of diversity of colony size. The three partial reversions obtained in the ultraviolet mutation experiments all have colony sizes greater than 2.0 mm, while all of the ultraviolet-induced phenotypic revertants with colony sizes of 1.3 mm or smaller proved to carry suppressor genes. Genetic tests with the selected spontaneous phenotypic revertants showed that

TABLE 3

Detection of suppressed mutants by a mixed-crossing technique

Crosses	T^- /colonies tested	Percent tryptophan requirers
Control:		
$cys^- \times td_2H^-$	21/68	30.9
Control:		
$cys^- \times$ mixture: td_2H^-, S-3*H*, S-14*H*, S-28*H* (equal parts)	13/156	8.3
Mixed cross of unknowns which do not segregate td_2:		
$cys^- \times$ mixture: S-13H^-, S-33H^-, S-32H^-, S-18H^- (equal parts)	0/156	0
Mixed cross of unknowns which segregate td_2:		
$cys^- \times$ mixture: S-1H^-, S-25H^-, S-19H^- (equal parts)	18/161	11.2
Subsequent single crosses to test S-1H^-, S-25H^-, S-19H^- *individually:*		
$cys^- \times$ S-1H^-	31/80	38.7
S-19H^-	34/68	50.0
S-25*H*	0/46	0

T^- = colonies requiring tryptophan.

all but three of the mutant strains tested were actually true reversions. These three carried suppressor genes and they produce colonies with a diameter of 1.7 or 1.8 mm (Figure 3). It therefore appears that suppressed strains are characteristically slow growers (yielding colonies with diameters less than 2.0 mm) while the true reversions form colonies varying in size from 0.5 mm to 4.5 mm.

Maintenance of mutant characteristics through transduction: Although the particular characteristics of indole accumulation and slow growth were maintained by each mutant in serial transfer, it was desirable to determine whether these same characteristics were maintained through transduction.

In several tests with suppressed mutants, it was shown that the suppressor gene

TABLE 4

Genetic data on selected spontaneous phenotypic revertants

Colony size	Genetic test	T^-/colonies tested	Percent tryptophan requirers
4.5, 4.4, 4.0, 4.0	Mixed Cross	0/141	0
3.7	Transduction	0/248	0
3.2	Transduction	0/36	0
3.2, 2.7, 2.7, 2.6	Mixed Cross	0/156	0
2.2	Transduction	0/49	0
2.1	Transduction	0/262	0
2.0	Transduction	0/120	0
1.8	Cross	69/146	47.3
1.8	Cross	0/587	0
1.7	Cross	155/301	51.5
1.7	Cross	74/124	59.7
1.5	Cross	0/100	0
1.5*, 1.4*, 1.3*	Mixed Cross	0/102	0
1.3	Cross	0/150	0
1.2	Cross	0/112	0
1.2	Transduction	0/87	0
0.56*	Mixed Cross	0/102	0
0.5	Transduction	0/61	0
	Control Transduction ($td_2 \rightarrow cys^-$)	168/493	34.1
	Control Cross ($cys^- \times td_2H^-$)	83/258	32.2

* These four strains were tested in the same mixed cross.
T^- = colonies requiring tryptophan.

TABLE 5

Genetic data on spontaneous phenotypic revertants unselected for colony size

No.	Colony size	Genetic test	T^-/colonies tested	Percent tryptophan requirers
1, 2, 3, 4, 5	Unknown	Mixed Cross	0/264	0
6, 7, 8, 9	Unknown	Mixed Cross	0/332	0
10, 11, 12, 13	Unknown	Mixed Cross	0/296	0
14, 15, 16, 17	Unknown	Mixed Cross	0/286	0

T^- = colonies requiring tryptophan.

could be transduced into a td_2 stock, yielding cultures which were indistinguishable from the original suppressed mutant. One experiment, for example, gave the following results when the control and transduction mixtures were plated on minimal agar: The td_2 reversion control gave three tryptophan-independent colonies per 5×10^7 cells plated, transduction of td_2 into td_2 gave one colony per 5×10^7 plated, while transduction of the two suppressor strains into td_2 gave 200 to 300 slow-growing colonies per 5×10^7 cells plated. In this experiment, twenty colonies were taken at random from each of the experimental plates and tested for indole accumulation. All of the colonies tested were indole accumulators and therefore the suppressor characteristics (slow growth on minimal agar and indole accumulation) were clearly transduced.

Although the frequency of mutation of the recipient strain to slow-growing indole accumulators was much lower than the transduction frequency of the suppressor gene, it was felt that studies on the inheritance of a particular growth rate through transduction should be performed on transduced mutants which could not possibly be mistakenly identified with new phenotypic revertants. When a true reversion is transduced into *cys*$^-$ it is recovered in the *cys*$^+$ progeny about 36 percent of the time. Four reversions (F, E, B, and H) were reisolated in this new genetic background (*cys*$^-$ strain) and their growth rates compared with the original revertant strains. A new wild type strain was also isolated from the products of one of these transductions. This strain was compared with the wild type K-12 strain which had been used as a control in other growth rate experiments. The value given for reversion E (before transduction) was obtained in a previous experiment.

It can be seen in Table 6 that, with the exception of strain H, the log phase generation time of the original strains and of the transduced strains agrees fairly

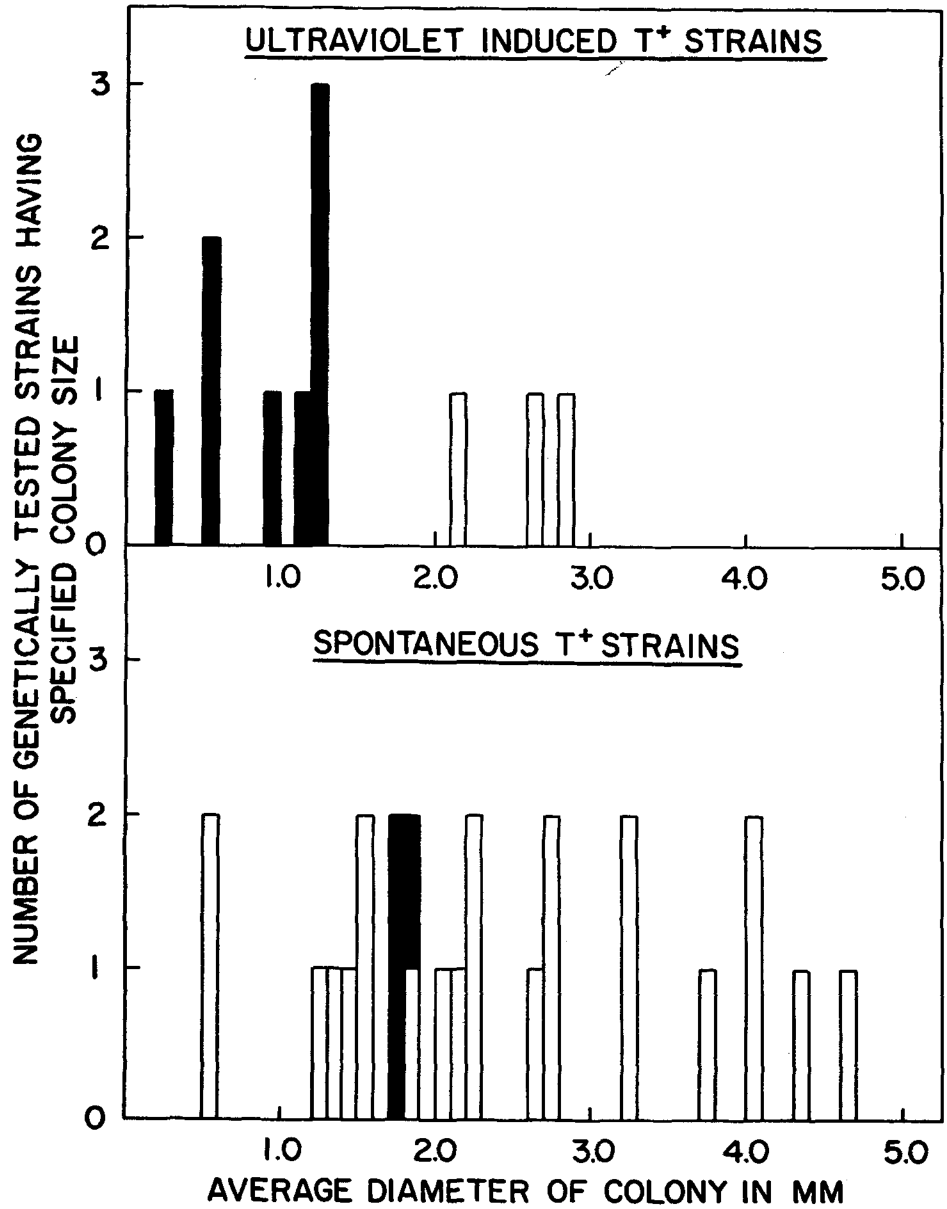


FIGURE 3.—Correlation of genetic type with average colony size. Black bars represent suppressed mutants and white bars, true revertants.

TABLE 6

Log phase generation time of true revertants before and after transduction into the cys⁻ *strain*

Stocks examined	Generation time before transduction*	Generation time after transduction
Reversion F	84′	87′
E	93′	102′
B	150′	165′
H	210′	189′
Prototroph recovered from F — *cys*⁻		66′
Wild type K-12	60′	

* = on unsupplemented minimal medium.

well. The small differences which were observed were not much greater than those obtained with the wild type control. The one exception, strain H, grows about ten percent faster after transduction than before. Since this strain is variable with respect to colony size, it is possible that a faster growing colony was picked from the transduction plate. In spite of this exception, the data on the other three reversions suggest that the growth rates characteristic of reversions are maintained through transduction. *Thus it would appear that the factor or factors controlling growth rate are associated with the form of the td locus which was transferred by transduction.*

Tryptophan synthetase activity in extracts of selected suppressed strains and revertants: To determine whether tryptophan synthetase activity is correlated with growth rate, revertants representing seven different colony size classes and suppressed strains representing two different colony size classes were selected for enzyme studies. Tryptophan synthetase activity was detected in the extracts of each phenotypic revertant examined (Table 7). However, there was marked

TABLE 7

The tryptophan synthetase specific activity of various suppressed mutants and revertants

Strain	Average diameter of colony	Generation time	Tryptophan synthetase specific activity
	mm.	minutes	
suppressed strain A*	0.5	192	0.57, 0.71
suppressed strain B	0.9	126	0.7
suppressed strain C	1.3	117	0.63, 0.71
suppressed strain D	1.3	135	0.5, 0.72
reversion A*	0.56	114	0.4
reversion B	1.2	123	0.3, 0.28
reversion C	2.2	111	0.8
reversion D*	2.7	96	1.9, 2.9
reversion E	3.2	93	0.8
reversion F	3.7	81	1.1, 1.25
reversion G	4.4	72	0.23, 0.26
wild type	5.0	69	2.9, 3.0

* = Some faster growing colonies were evident on the control platings of these strains.

variation in the level of enzyme activity. When comparisons of activity of the extractable tryptophan synthetase are made between extracts of any phenotypic revertants, the reliability of the procedure employed must be considered in detail. As stated earlier, cultures were harvested for extraction when the concentration of cells was somewhere between 5×10^8/ml and 10^9/ml. One revertant was grown many times and repeatedly sampled at levels of growth corresponding to 5×10^8 cells/ml and 10^9 cells/ml. The variation between the activities of these extracts (roughly ten percent) is no more than the variation observed between extracts from cells grown at different times and sampled at the same turbidity. In all cases colony size controls were performed on the inoculum and on the cells at the time of harvesting. With the exception of reversions A and D (as noted in Table 7) no pronounced selection for faster growing types occurred.

The four selected suppressed strains have tryptophan synthetase specific activities only about 20–25 percent that of wild type K-12 (Table 7). Although differences in growth rate occur among the four suppressed stocks, this is certainly not reflected in the tryptophan synthetase activity of their extracts.

The seven selected revertants (Table 7) show much more variation in their specific activities—from ten percent of wild type activity to values approaching the wild type range. Enzyme content is not, however, strictly correlated with colony size. Reversion G grows well on minimal agar (average colony diameter, 4.4 mm) but has only ten percent of the wild type tryptophan synthetase activity. Conversely, reversion D grows more poorly (average colony diameter, 2.7 mm) but has high tryptophan synthetase specific activity. However, the control plates in the reversion D experiments showed evidence of the appearance of faster growers during the experiment. Colonies on these plates varied from 2.0 mm to 3.3 mm in diameter. If the colonies of 3.3 mm represented the fastest growers in the population, this strain still shows more tryptophan synthetase activity than would be expected in comparison with the other reversions.

If reversions D and G are omitted from the comparisons, there is a very rough positive correlation between growth rate and specific activity.

All of the extracts examined were frozen after sonic treatment and tryptophan synthetase assays were usually performed within one to two days after the preparation of the extract. On a few occasions, however, an extract was reassayed a week or more after the original determination had been made. In three of these cases (reversions B, F, and G) the enzymatic activity had dropped considerably. The tryptophan synthetase activity in extracts of wild type K-12 remains stable under these conditions for as long as six months.

Genetic examination of the site of mutation in reversions: The fact that many of the revertants examined differed considerably from wild type in growth rate, accumulation behavior, and enzyme content raised the possibility that the second mutation, although apparently occurring within the *td* locus, may have occurred at a site distinct from the site of the original mutation. If this were the case, it would be expected that the original mutant area could be reisolated by a rare crossover event within the *td* locus.

With this in mind, crosses similar to those previously described were performed on a larger scale. Four histidine-requiring td_2 reversions, A, D, G, and I, were crossed with cys^- td^+ and plated in the presence of tryptophan. The cys^+ recombinants obtained were tested for their tryptophan requirement (Table 8).

TABLE 8

Tests for a rare crossover event in several revertants

Reversion*	Genetic test	T^-/colonies tested
A	Cross: $cys^- \times AH^-$	0/421
D	Cross: $cys^- \times DH^-$	0/860
G	Cross: $cys^- \times GH^-$	0/1308
I	Cross: $cys^- \times IH^-$	0/819

* = The four reversions tested have widely different characteristic colony sizes.
T^- = colonies requiring tryptophan.

In analyzing the results of these crosses, two alternative possibilities must be considered: first, the site of reversion may be to the right of the td_2 mutant area (see diagram 1), and second, the site of reversion may be between cys^- and the td_2 mutant area (see diagram 2).

Diagram (1)

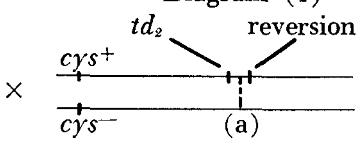
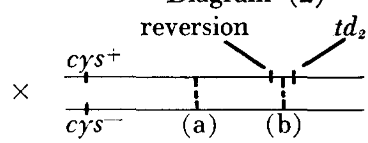


Diagram (2)



If model (1) describes the situation, a rare single crossover at (a) would give a td_2 type in the cys^+ progeny. If model (2) is correct, a double crossover (a) (b) would be required to give a cys^+ td_2. However, in bacteria, it is probable that only fragments of the donor chromosomes participate in recombination. In this case a double crossover in (1) and a quadruple crossover in (2) would be required to give cys^+ td_2.

Large numbers of cys^+ progeny were tested and no T^- (cys^+ td_2) types were recovered. Since none were recovered, either the second mutation occurred at or extremely close to the site of the original mutation, or model (2) may represent the true situation in these particular strains.

DISCUSSION

Tryptophan-independent phenotypic revertants obtained from the tryptophan-requiring mutant td_2 arise by both suppressor mutation and reversion. More than half of the phenotypic revertants which are recovered accumulate indole even though these same strains are able to grow in the absence of exogenously supplied tryptophan. This is true for both suppressed strains and true revertants. Of particular interest is the fact that in many of the strains examined wild type physiological activity was not quantitatively restored by reversion at the original mutant locus. The name *partial reversion* will be used to describe this situation. Furthermore, the partial reversion class has been shown to include a striking variety of revertants that can be distinguished by characteristic growth rates and levels of enzymatic activity.

Although, as expected, the number of phenotypic revertants increased after ultraviolet irradiation, the distribution of colony sizes was different from that observed in spontaneous mutation experiments. The spontaneous phenotypic revertants generally were of many different colony size classes with each of these classes including about the same number of strains. However, following ultraviolet treatment, proportionately more colonies less than 2.0 mm in diameter were obtained. Genetic studies have shown that these slower growing ultraviolet induced phenotypic revertants are suppressed mutants while the faster growing types are partial revertants. In the spontaneous mutation experiments, most of the T^+ strains obtained were partial revertants. The few suppressed mutants which were isolated produced colonies 1.7–1.8 mm. in diameter.

Considering these facts, it would seem that the ultraviolet treatment produces more suppressor mutations than true reversions. Although negative selection might explain the relative rarity of appearance of suppressed strains in the spontaneous mutation experiments, in a recovery experiment which was performed this was not the case. The explanation for the observed difference is not known.

The enzyme studies shows that tryptophan synthetase activity has been restored to some extent in both the suppressed mutants and the partial revertants. The strains chosen for these enzyme studies have widely different growth rates and it was hoped that, within each genetic group, there would be some correlation of growth rate with enzymatic activity. Such a correlation would be expected if tryptophan synthetase activity were the only factor limiting growth. The fact that all of these strains exhibit a growth response to added tryptophan demonstrates that the synthesis of tryptophan is one of the growth limiting factors. The four suppressed stocks which were examined showed no correlation between growth rate and enzyme content. Thus the full effect of suppressor mutations cannot be measured in terms of the amount of extractable enzyme present in crude preparations. With the partial reversions, there is some suggestion of a very rough correlation between growth rate and the amount of extractable enzyme activity; however, one of the partial reversions examined has much less tryptophan synthetase than would be expected and one has much more. Certainly for reversion D, which grows almost as well as wild type but has only ten percent of the wild type enzyme activity, it can be said that the physiological effect of the reverse mutation is not reflected in the quantity of extractable tryptophan synthetase which was detected.

In view of the poor correlation in some cases between extractable enzyme levels and growth rate and the preliminary observation that the tryptophan synthetase in some of the partial revertants may be relatively labile on storage, it might be postulated that all of these partial reversions represent qualitative changes in tryptophan synthetase, i.e., that mutation has restored the ability to form tryptophan synthetase but the enzyme formed is not identical with the wild type enzyme. Similar observations of instability or of alteration of specific enzymes restored by reversions have been reported by GILES (1957) and by FINCHAM (1957) in studies with Neurospora.

It was mentioned previously that, in addition to the indole accumulating phenotypic revertants, nonaccumulators were also obtained. These were originally believed to represent complete reversion to wild type rather than partial reversion. Preliminary enzyme and growth studies, however, indicate that these nonaccumulators do not exhibit wild type characteristics although those strains which were examined were found to be true revertants by genetic tests.

In addition it should be emphasized that although the mutational behavior described in this paper is characteristic for strain td_2, an allele (td_4) of this mutant seems to behave quite differently. Preliminary investigations with td_4 suggest that it always reverts to a nonindole accumulating, fast-growing type.

Preliminary genetic tests were performed with the partial revertants of strain td_2 in an attempt to separate the sites of the original and second mutations. No evidence of separability was obtained. However, in view of the fact that many distinguishable partial revertants, each possessing tryptophan synthetase activity, are recovered from a single mutant, it would appear unlikely that all of the reversions occurred at the exact site of the original mutation. Rather, it is more probable that mutational changes at other sites within the same gene can result in the formation of tryptophan synthetase proteins with characteristically different turnover numbers.

SUMMARY

1. Tryptophan-independent (T^+) phenotypic revertants were obtained from td_2, a tryptophan-requiring mutant of *E. coli* strain K-12, both spontaneously and following ultraviolet treatment.

2. Genetic studies show that T^+ phenotypic revertants, either spontaneous or ultraviolet induced, arise both by suppressor mutation and reversion. Proportionately more suppressor mutations occur following ultraviolet treatment.

3. Growth rate studies demonstrate that suppressed mutants are characteristically slow growers, while true revertants have widely different growth rates only sometimes approaching the wild type level. Revertants of this type are called partial revertants.

4. The enzyme studies performed show that tryptophan synthetase activity has been restored to some extent in both suppressed mutants and partial revertants. However, studies of suppressed mutants and partial revertants with widely different growth rates show that in most cases enzymatic activity probably cannot be correlated with growth rate. Also, the extracted tryptophan synthetase of three partial revertants is more labile on storage than is the wild type enzyme. In view of these two latter observations it is postulated that in partial reversion the back mutation has restored the ability to form tryptophan synthetase but that the enzyme may not be qualitatively comparable to the wild type enzyme.

5. Preliminary large-scale genetic tests with four partial revertants have failed to separate the sites of the original and the second mutation. However, considering the fact that many distinguishably different partial revertants, all with tryptophan synthetase activity, are recovered from a single mutant it is proposed that some of the second mutational changes may have occurred at other sites within the original mutant gene.

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